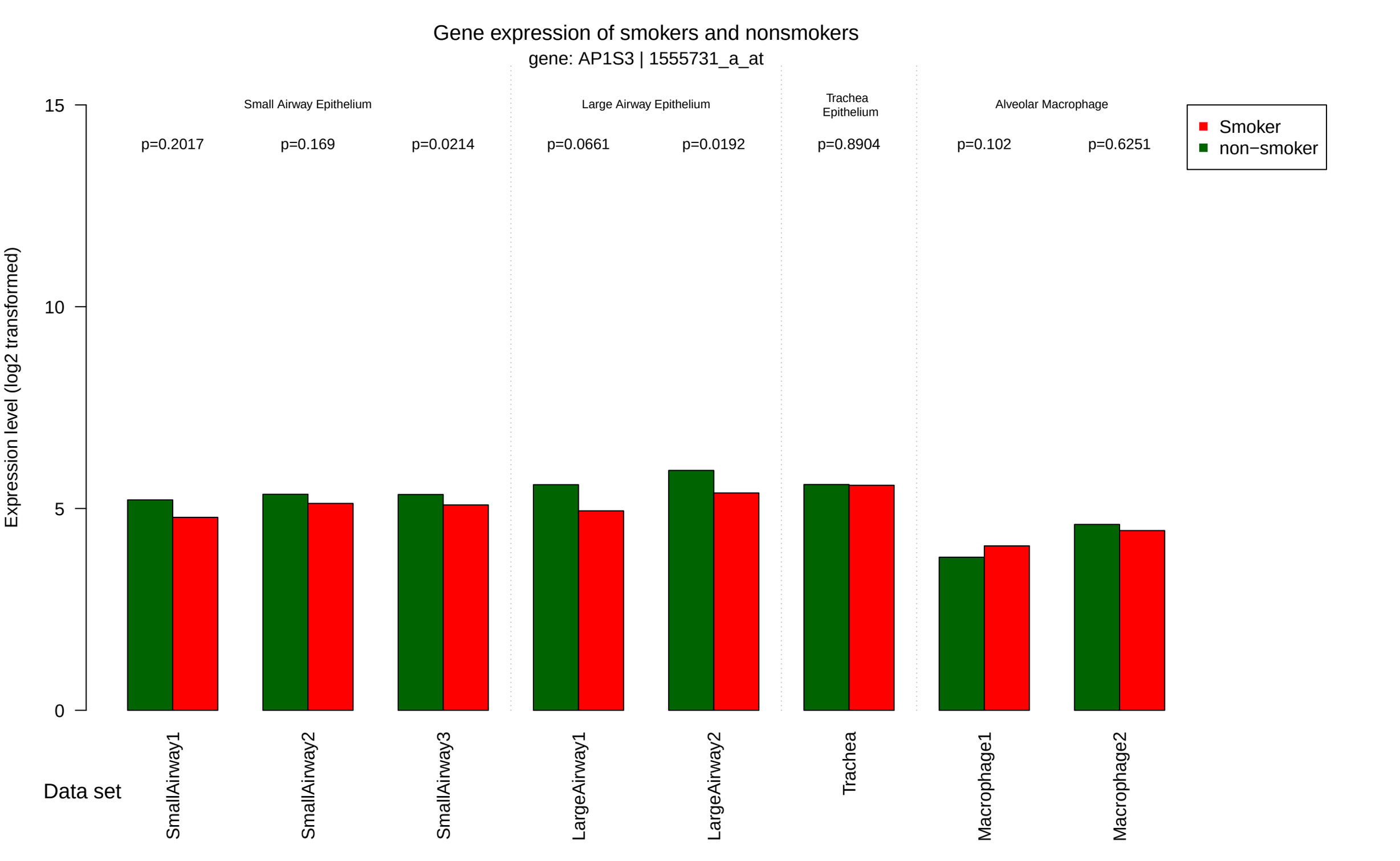
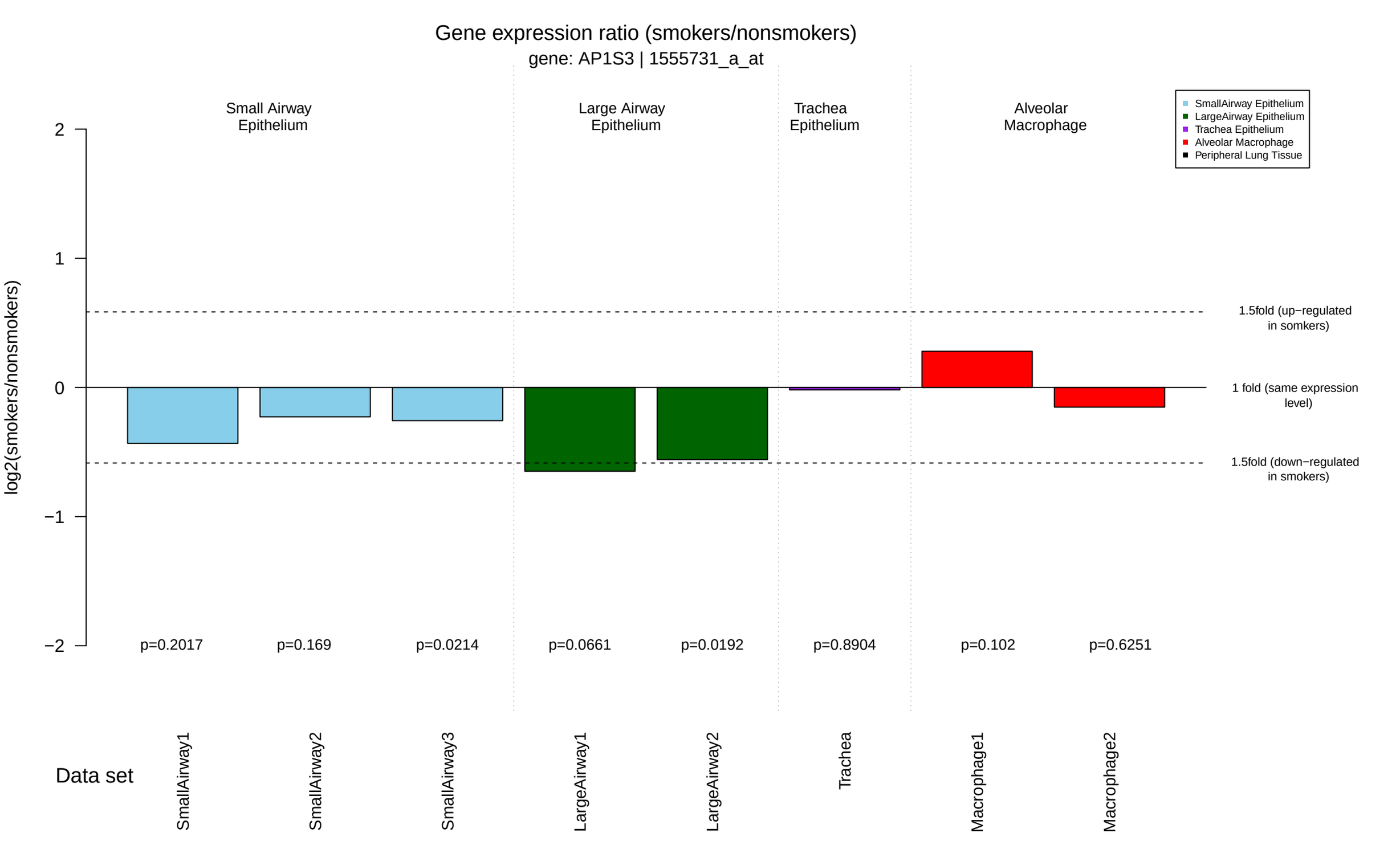
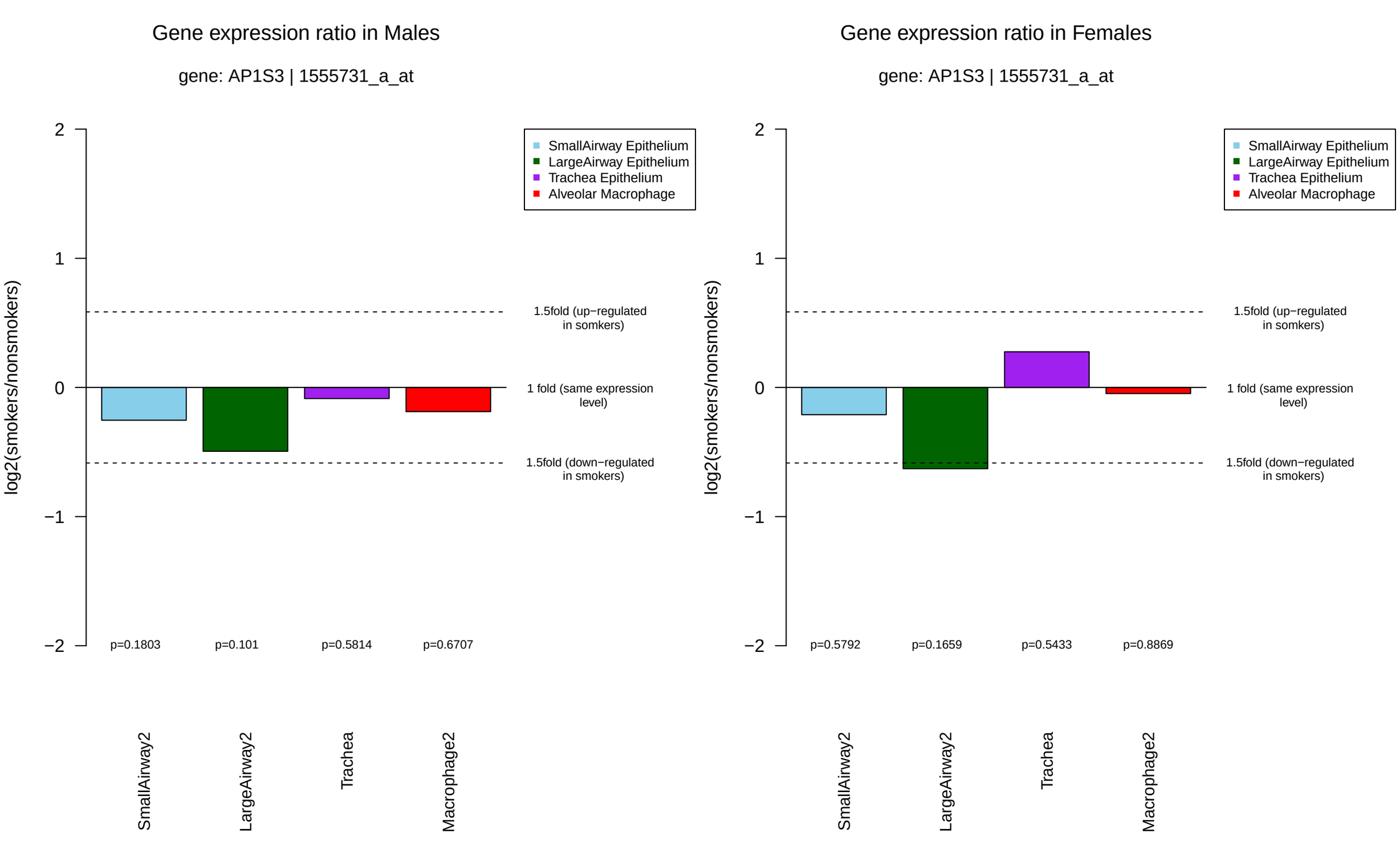
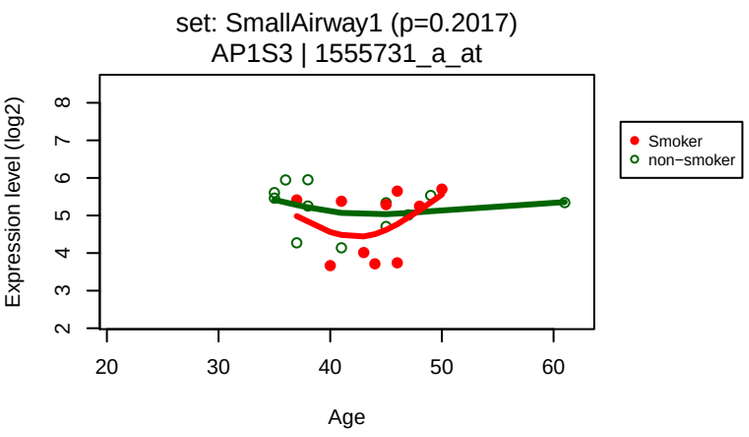
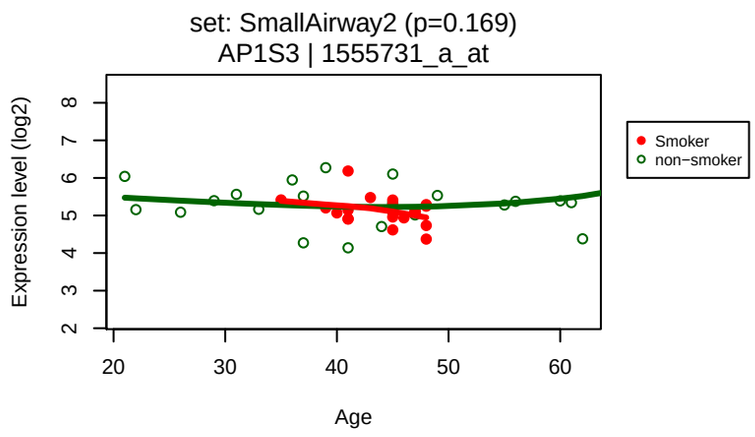
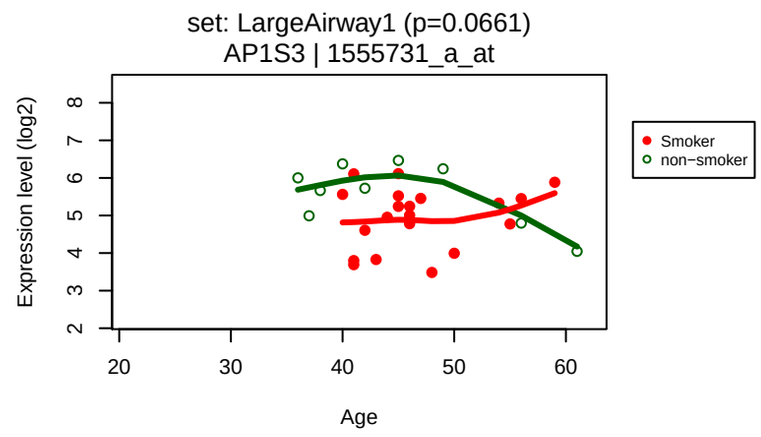

Gene expression of smokers and nonsmokers
gene: AP1S3 | 1555731_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2017
p=0.169
p=0.0214
p=0.0661
p=0.0192
p=0.8904
p=0.102
p=0.6251
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

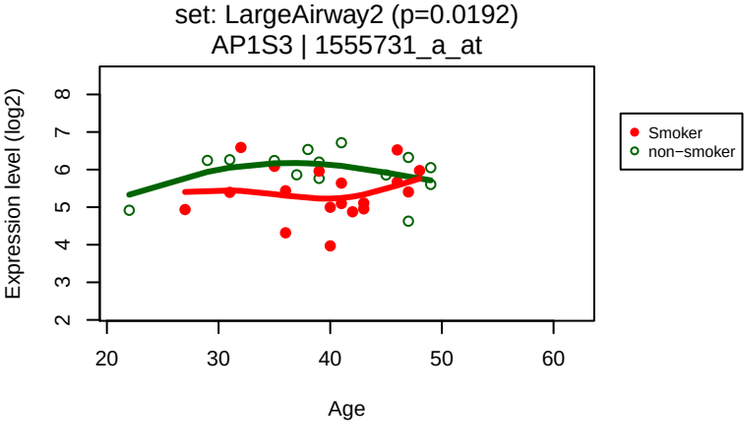

Gene expression ratio (smokers/nonsmokers)
gene: AP1S3 | 1555731_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2017
p=0.169
p=0.0214
p=0.0661
p=0.0192
p=0.8904
p=0.102
p=0.6251
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

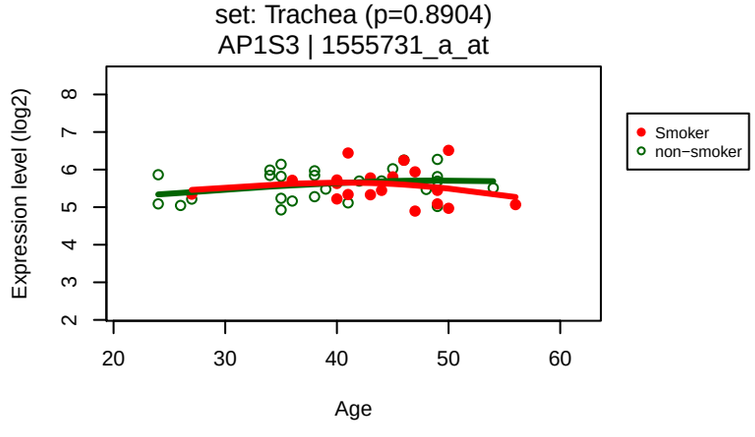

Gene expression ratio in Males
gene: AP1S3 | 1555731_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1803
p=0.101
p=0.5814
p=0.6707
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP1S3 | 1555731_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5792
p=0.1659
p=0.5433
p=0.8869
SmallAirway2
LargeAirway2
Trachea
Macrophage2

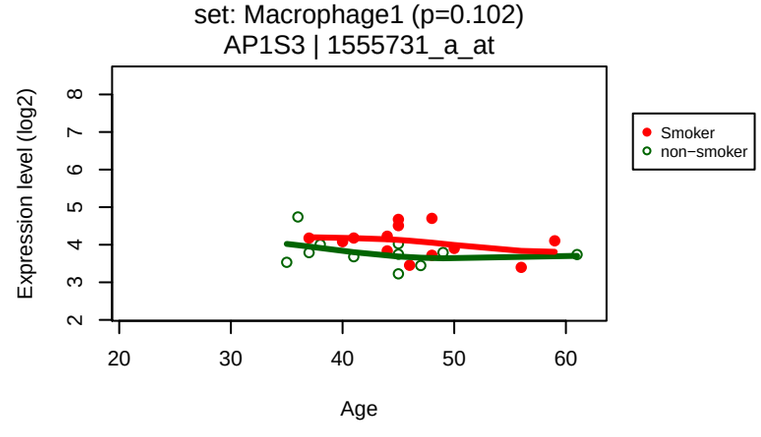

set: SmallAirway1 (p=0.2017)
AP1S3 | 1555731_a_at
Expression level (log2)
Age
Smoker
non−smoker

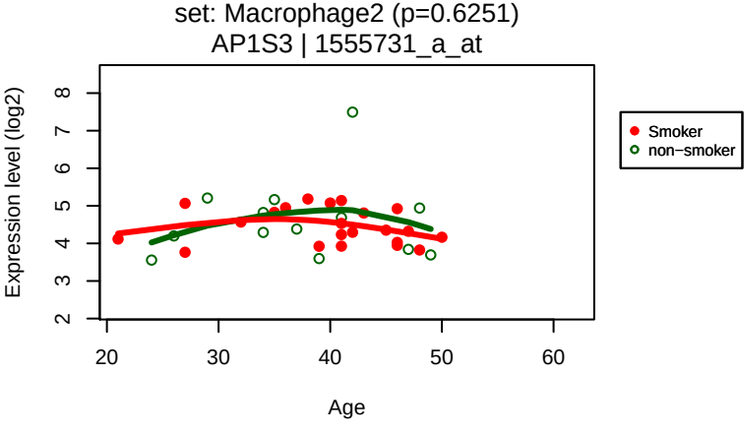

set: SmallAirway2 (p=0.169)
AP1S3 | 1555731_a_at
Expression level (log2)
Age
Smoker
non−smoker

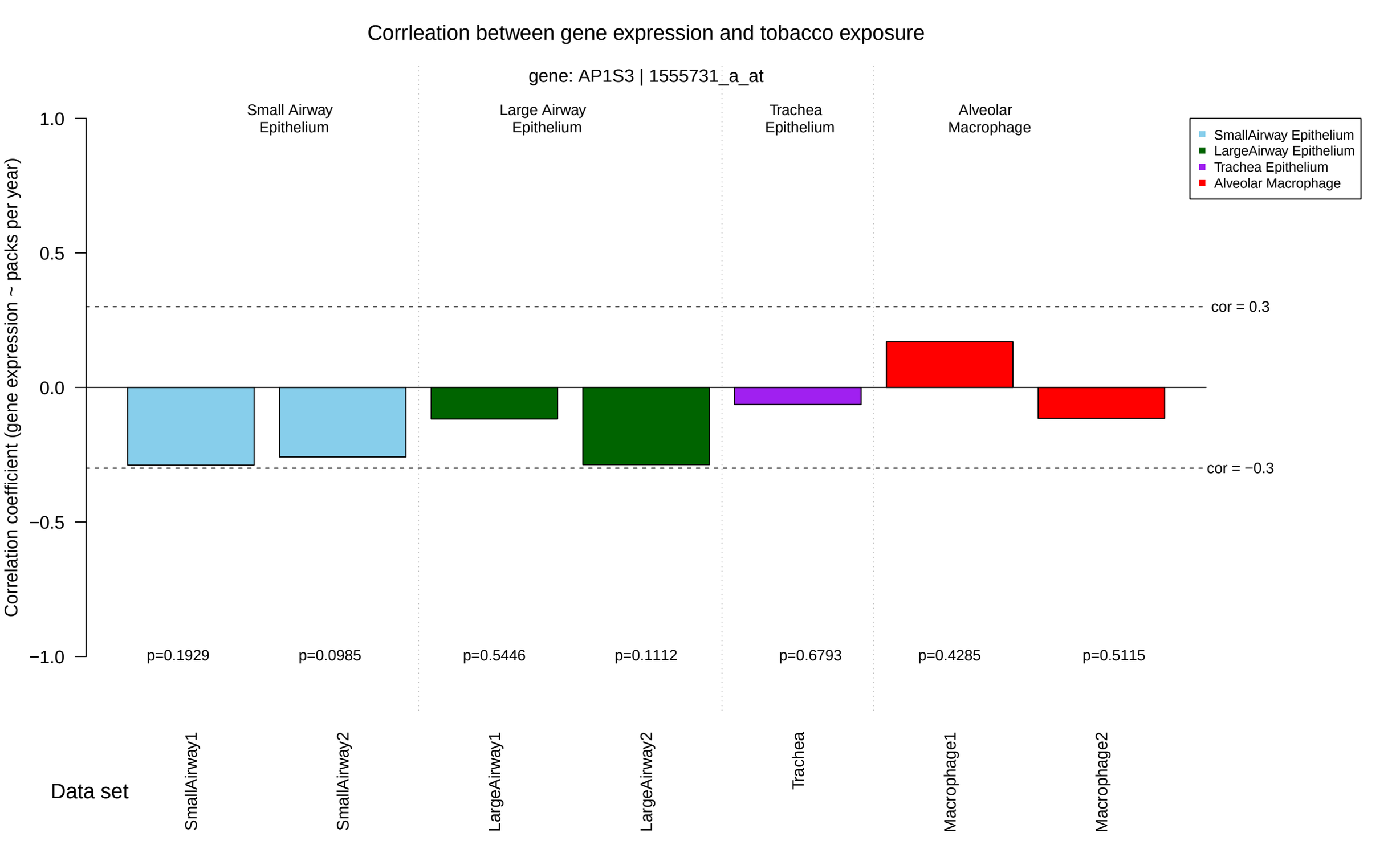

set: LargeAirway1 (p=0.0661)
AP1S3 | 1555731_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0192)
AP1S3 | 1555731_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8904)
AP1S3 | 1555731_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.102)
AP1S3 | 1555731_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6251)
AP1S3 | 1555731_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP1S3 | 1555731_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1929
p=0.0985
p=0.5446
p=0.1112
p=0.6793
p=0.4285
p=0.5115
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2